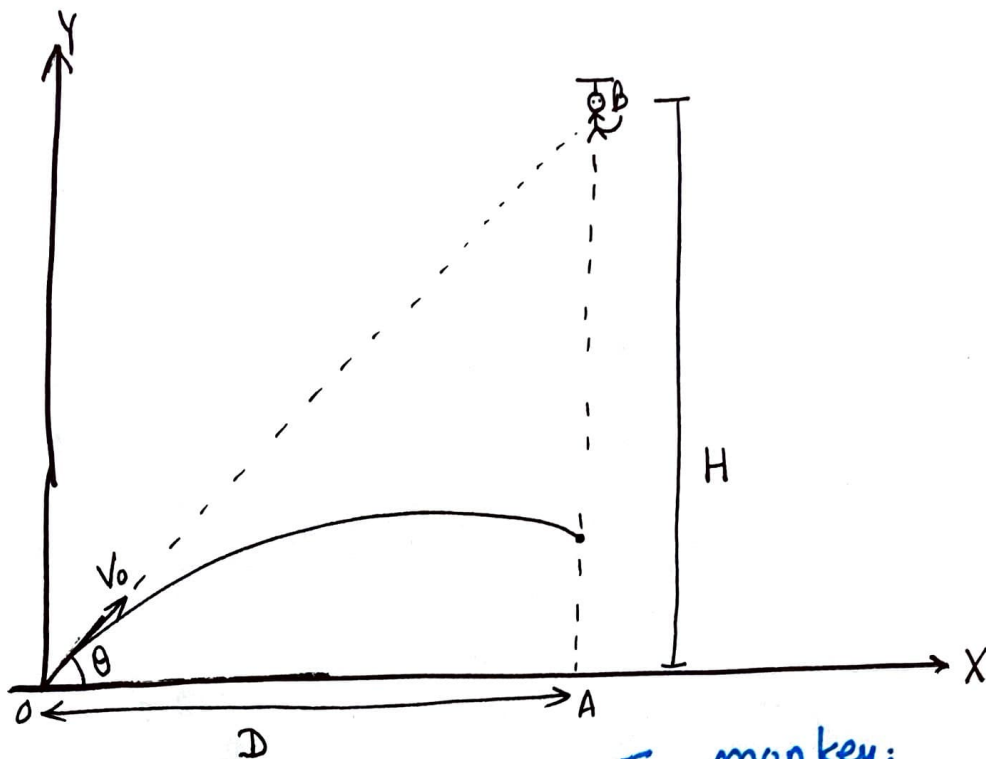




For bullet:

$$X(t) = V_0 \cos \theta t$$

$$Y(t) = V_0 \sin \theta t - \frac{1}{2} g t^2$$

For monkey:

$$X(t) = D$$

$$y(t) = H - \frac{1}{2} g t^2$$

~~Mon~~ Bullet will hit monkey if,

$$X(t) = X(t) \\ V_0 \cos \theta t = D \quad \text{--- (i)} \Rightarrow V_0 t = \frac{D}{\cos \theta}$$

And; $Y(t) = y(t)$

$$V_0 \sin \theta t - \frac{1}{2} g t^2 = H - \frac{1}{2} g t^2$$

$$V_0 \sin \theta t = H$$

$$V_0 t \cdot \sin \theta = H$$

$$\frac{D}{\cos \theta} \sin \theta = H$$

$\tan \theta = \frac{H}{D}$ which is true for larger triangle OAB. So, monkey will be killed by the bullet.

So $y(t) (\text{monkey}) = y(t) (\text{bullet})$